



Multi-branch fiber comb with relative frequency uncertainty at 10^{-20} using fiber noise difference cancellation

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Abstract: In multi-branch combs, the comb outputs from the branches suffer from different fiber noises, which often limit the uncertainty of the combs referring a highly-stable optical frequency. To overcome this limitation, we introduced fiber noise difference cancellation to multi-branch fiber combs. We detected and phase-locked the beat notes between the branch outputs and a common 1542 nm continuous wave laser. A piezo-electric transducer-based fiber stretcher was installed in each branch except for the branch used as the cancellation reference. We fabricated two quasi-identical combs with this mechanism and confirmed the relative frequency uncertainty by comparing them. The cancellation improved the frequency uncertainty to a low level of 10^{-20} at a 100000-s averaging time.

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References and links

1. T. Udem, J. Reichert, R. Holzwarth, and T. W. Hänsch, "Accurate measurement of large optical frequency differences with a mode-locked laser," *Opt. Lett.* **24**(13), 881–883 (1999).
2. D. J. Jones, S. A. Diddams, J. K. Ranka, A. Stentz, R. S. Windeler, J. L. Hall, and S. T. Cundiff, "Carrier-envelope phase control of femtosecond mode-locked lasers and direct optical frequency synthesis," *Science* **288**(5466), 635–639 (2000).
3. J. Stenger, H. Schnatz, C. Tamm, and H. R. Telle, "Ultraprecise measurement of optical frequency ratios," *Phys. Rev. Lett.* **88**(7), 073601 (2002).
4. L.-S. Ma, Z. Bi, A. Bartels, L. Robertsson, M. Zucco, R. S. Windeler, G. Wilpers, C. Oates, L. Hollberg, and S. A. Diddams, "Optical frequency synthesis and comparison with uncertainty at the 10^{-19} level," *Science* **303**(5665), 1843–1845 (2004).
5. A. Bartels, C. W. Oates, L. Hollberg, and S. A. Diddams, "Stabilization of femtosecond laser frequency combs with subhertz residual linewidths," *Opt. Lett.* **29**(10), 1081–1083 (2004).
6. W. C. Swann, J. J. McFerran, I. Coddington, N. R. Newbury, I. Hartl, M. E. Fermann, P. S. Westbrook, J. W. Nicholson, K. S. Feder, C. Langrock, and M. M. Fejer, "Fiber-laser frequency combs with subhertz relative linewidths," *Opt. Lett.* **31**(20), 3046–3048 (2006).
7. T. R. Schibli, I. Hartl, D. C. Yost, M. J. Martin, A. Marcinkevičius, M. E. Fermann, and J. Ye, "Optical frequency comb with submillihertz linewidth and more than 10 W average power," *Nat. Photonics* **2**(6), 355–359 (2008).
8. M. J. Martin, S. M. Foreman, T. R. Schibli, and J. Ye, "Testing ultrafast mode-locking at microhertz relative optical linewidth," *Opt. Express* **17**(2), 558–568 (2009).
9. T. Legero, C. Lisdat, J. S. R. Vellore Winfred, H. Schnatz, G. Grosche, F. Riehle, and U. Sterr, "Interrogation laser for a strontium lattice clock," *IEEE Trans. Instrum. Meas.* **58**(4), 1252–1257 (2009).
10. Y. Nakajima, H. Inaba, K. Hosaka, K. Minoshima, A. Onae, M. Yasuda, T. Kohno, S. Kawato, T. Kobayashi, T. Katsuyama, and F.-L. Hong, "A multi-branch, fiber-based frequency comb with millihertz-level relative linewidths using an intra-cavity electro-optic modulator," *Opt. Express* **18**(2), 1667–1676 (2010).
11. H. Inaba, K. Hosaka, M. Yasuda, Y. Nakajima, K. Iwakuni, D. Akamatsu, S. Okubo, T. Kohno, A. Onae, and F.-L. Hong, "Spectroscopy of ^{171}Yb in an optical lattice based on laser linewidth transfer using a narrow linewidth frequency comb," *Opt. Express* **21**(7), 7891–7896 (2013).

12. K. Iwakuni, H. Inaba, Y. Nakajima, T. Kobayashi, K. Hosaka, A. Onae, and F.-L. Hong, "Narrow linewidth comb realized with a mode-locked fiber laser using an intra-cavity waveguide electro-optic modulator for high-speed control," *Opt. Express* **20**(13), 13769–13776 (2012).
13. D. Nicolodi, B. Argence, W. Zhang, R. Le Targat, G. Santarelli, and Y. Le Coq, "Spectral purity transfer between optical wavelengths at the 10^{-18} level," *Nat. Photonics* **8**(3), 219–223 (2014).
14. A. Apolonski, A. Poppe, G. Tempea, C. Spielmann, T. Udem, R. Holzwarth, T. W. Hänsch, and F. Krausz, "Controlling the phase evolution of few-cycle light pulses," *Phys. Rev. Lett.* **85**(4), 740–743 (2000).
15. F. Tauser, A. Leitenstorfer, and W. Zinth, "Amplified femtosecond pulses from an Er:fiber system: Nonlinear pulse shortening and selfreferencing detection of the carrier-envelope phase evolution," *Opt. Express* **11**(6), 594–600 (2003).
16. T. R. Schibli, K. Minoshima, F.-L. Hong, H. Inaba, A. Onae, H. Matsumoto, I. Hartl, and M. E. Fermann, "Frequency metrology with a turnkey all-fiber system," *Opt. Lett.* **29**(21), 2467–2469 (2004).
17. F.-L. Hong, K. Minoshima, A. Onae, H. Inaba, H. Takada, A. Hirai, H. Matsumoto, T. Sugiura, and M. Yoshida, "Broad-spectrum frequency comb generation and carrier-envelope offset frequency measurement by second-harmonic generation of a mode-locked fiber laser," *Opt. Lett.* **28**(17), 1516–1518 (2003).
18. B. R. Washburn, R. W. Fox, N. R. Newbury, J. W. Nicholson, K. Feder, P. S. Westbrook, and C. G. Jørgensen, "Fiber-laser-based frequency comb with a tunable repetition rate," *Opt. Express* **12**(20), 4999–5004 (2004).
19. P. Pal, W. H. Knox, I. Hartl, and M. E. Fermann, "Self referenced Yb-fiber-laser frequency comb using a dispersion micromanaged tapered holey fiber," *Opt. Express* **15**(19), 12161–12166 (2007).
20. C. Benko, A. Ruehl, M. J. Martin, K. S. E. Eikema, M. E. Fermann, I. Hartl, and J. Ye, "Full phase stabilization of a Yb:fiber femtosecond frequency comb via high-bandwidth transducers," *Opt. Lett.* **37**(12), 2196–2198 (2012).
21. T. Nakamura, I. Ito, and Y. Kobayashi, "Offset-free broadband Yb:fiber optical frequency comb for optical clocks," *Opt. Express* **23**(15), 19376–19381 (2015).
22. W. Zhang, M. Lours, M. Fischer, R. Holzwarth, G. Santarelli, and Y. Coq, "Characterizing a fiber-based frequency comb with electro-optic modulator," *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **59**(3), 432–438 (2012).
23. C. Hagemann, C. Grebing, T. Kessler, S. Falke, N. Lemke, C. Lisdat, H. Schnatz, F. Riehle, and U. Sterr, "Providing 10^{-16} short-term stability of a 1.5 μm laser to optical clocks," *IEEE Trans. Instrum. Meas.* **62**(6), 1556–1562 (2013).
24. N. Ohmae, N. Kuse, M. E. Fermann, and H. Katori, "All-polarization-maintaining, single-port Er:fiber comb for high-stability comparison of optical lattice clocks," *Appl. Phys. Express* **10**(6), 062503 (2017).
25. H. Leopardi, J. Davila-Rodriguez, F. Quinlan, J. Olson, J. A. Sherman, S. A. Diddams, and T. M. Fortier, "Single-branch Er:fiber frequency comb for precision optical metrology with 10^{-18} fractional instability," *Optica* **4**(8), 879–885 (2017).
26. Y. Zhang, S. Fan, L. Yan, L. Zhang, X. Zhang, W. Guo, S. Zhang, and H. Jiang, "Robust optical-frequency-comb based on the hybrid mode-locked Er:fiber femtosecond laser," *Opt. Express* **25**(18), 21719–21725 (2017).
27. L. A. M. Johnson, P. Gill, and H. S. Margolis, "Evaluating the performance of the NPL femtosecond frequency combs: agreement at the 10^{-21} level," *Metrologia* **52**(1), 62–71 (2015).
28. Y. Yao, Y. Jiang, H. Yu, Z. Bi, and L. Ma, "Optical frequency divider with division uncertainty at the 10^{-21} level," *Natl. Sci. Rev.* **3**, 463–469 (2016).
29. Y. Nakajima, H. Inaba, F.-L. Hong, A. Onae, K. Minoshima, T. Kobayashi, M. Nakazawa, and H. Matsumoto, "Optimized amplification of femtosecond optical pulses by dispersion management for octave-spanning optical frequency comb generation," *Opt. Commun.* **281**(17), 4484–4487 (2008).
30. M. Hirano, T. Nakanishi, T. Okuno, and M. Onishi, "Silica-based highly nonlinear fibers and their application," *IEEE J. Sel. Top. Quantum Electron.* **15**, 103–113 (2009).
31. K. Hosaka, H. Inaba, Y. Nakajima, M. Yasuda, T. Kohno, A. Onae, and F.-L. Hong, "Evaluation of the clock laser for an Yb lattice clock using an optic fiber comb," *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **57**(3), 606–612 (2010).
32. L. S. Ma, P. Jungner, J. Ye, and J. L. Hall, "Delivering the same optical frequency at two places: accurate cancellation of phase noise introduced by an optical fiber or other time-varying path," *Opt. Lett.* **19**(21), 1777–1779 (1994).
33. P. A. Williams, W. C. Swann, and N. R. Newbury, "High-stability transfer of an optical frequency over long fiber-optic links," *J. Opt. Soc. Am. B* **25**(8), 1284–1293 (2008).

1. Introduction

The invention and development of optical frequency combs made it possible to coherently link frequencies in the microwave and optical domains [1,2] and also to transfer even the spectral linewidth of optical frequency references to other wavelengths exceeding the octave range [3–13]. Optical frequency combs are usually generated from mode-locked lasers (oscillators) and have equally spaced modes in the optical frequency domain. Several oscillators among the various available types have had great success and become mainstream devices. For example, Ti:sapphire based combs [1,2,4,5,8,9,14], erbium-fiber based combs [6,9,15–18], and ytterbium-fiber based combs [7,19–21] are familiar to the community. Of the diverse types of

combs, erbium-fiber based combs have been widely used because of their robust all-fiber design.

Improvement in the frequency stability of combs has been intensively studied mainly for optical clocks [4,10,12,13]. In these studies, the frequency stability has reached $2\text{--}3 \times 10^{-19}$ for an averaging time of more than 10000-s. Such high stabilities were achieved by locking the carrier-envelope-offset frequency (f_{CEO}) and the beat note frequency (f_{beat}) between a comb mode and an ultrastable single frequency continuous wave (CW) reference laser, without considering the repetition frequency (f_{rep}) for comb stabilization. In particular, comb-stabilization based on optical references provided by an actuator with a wide control bandwidth, such as an intracavity electro-optic modulator (EOM), can realize a high signal to noise ratio for a coherent peak with low phase noise [10–12,22].

These studies commonly employed multi-branch/path configurations. An oscillator output is divided into certain branches/paths to allow the respective outputs to be used for different applications. The inter-branch/path phase noise differences degrade the frequency stabilities [23]. In other words, although the feedback control of the comb stabilization (for f_{CEO} or f_{beat}) stabilizes the comb frequency including the phase noise originating from the branch paths, another branch output suffers from not only the phase noise from the branch paths but also from the phase noise from the branch used for comb stabilization. Therefore, single branch/path configurations were employed to avoid inter-branch phase noise differences and this resulted in a frequency stability at the 10^{-21} – 10^{-20} level for long averaging times [24–28]. However, we believe that fiber combs with multiple outputs (multi-branch fiber combs) still have diverse advantages including the spectral tailoring of a nonlinearly broadened spectrum and a high signal to noise ratio of the beat note with a CW laser output.

In this study, we report a multi-branch fiber-based frequency comb with a relative frequency uncertainty at the 10^{-20} level. We employed piezo-electric-transducer (PZT)-based fiber stretchers to cancel the fiber noise differences between the multiple branches. To evaluate the cancellation effect and the relative frequency uncertainty, we compared two of the branch output frequencies in the optical frequency domain with another quasi-identical comb.

2. Experimental setup

2.1 Multi-branch comb configuration

Figure 1 shows a schematic diagram of our experimental setup for comb generation in a multi-branch configuration and inter-branch fiber noise difference cancellation (FNDC). The oscillator was an erbium-doped fiber laser that was mode-locked by nonlinear polarization rotation. Its average power, pulse width, center wavelength and repetition frequency were 4.4 mW, 135 fs, 1560 nm, and 43.4 MHz, respectively. The oscillator output pulse was divided into four branches. (One branch was not used in this study.) Each divided pulse was optimally amplified by an erbium-doped fiber amplifier [29] and was spectrally broadened by a subsequent highly nonlinear fiber (HNLF) [30]. In the first branch (referred as the “CEO-locking branch”), a portion of the HNLF output was guided to an $f\text{--}2f$ interferometer to detect the CEO beat of the comb. f_{CEO} was phase-locked to an RF reference at 30 MHz by feedback control to the injection current of the pump LD of the oscillator. In the second branch (referred as the “beat-locking branch”), a beat signal between an ultrastable reference laser at 1064 nm [31] and the nearest comb line was locked to an RF reference to realize an f_{beat} of 30 MHz. The fast and slow servos feedback controlled a waveguide-type EOM in the oscillator cavity and a temperature controller that controlled the cavity temperature, respectively. The comb generation and the locking configuration explained above are described in detail in our previous report [12]. The output from the third branch (referred as the “application branch”), which was outside of the feedback loop for the comb stabilization, was used for comparison with another comb. In the beat-locking branch, a beat signal between a comb mode and a CW

laser emitting at 1542 nm, which is explained in the following paragraph, is also used for another comparison. The comparison procedure is explained later in detail in subsection 2.2.

As shown in Fig. 1, f_{CEO} was detected through the CEO-locking branch, and fiber noise caused in this branch was cancelled by the feedback control to the pump LD. The outputs from the other two branches suffered from different fiber noises that originated from the fiber paths, which were not common to the CEO-locking branch. To suppress fluctuation due to the fiber noise difference, we employed inter-branch FNDC in the beat-locking and application branches. The FNDC is similar to conventional fiber noise cancellation [32]. In each of the two branches, we installed an actuator for the FNDC, which was a fiber stretcher based on a drum-type PZT. The FNDC cancelled out the fiber noise differences between the CEO-locking branch and the other two branches. A beat note between the CW laser output at 1542 nm and the nearest comb mode of the CEO-locking branch output was used as a reference signal for the FNDC. The laser was loosely locked to the comb by using the reference signal to stably obtain the signal. Beat note signals between the CW laser and the outputs from the beat-locking and application branches were also detected and were used as error signals for the FNDC in the respective branches. The fiber stretchers installed in the two branches controlled the fiber lengths of the branches and cancelled out the phase differences between the reference and the respective error signals. Thus, the residual fiber noise was the same in each branch. Finally, the feedback control of the f_{CEO} and the f_{beat} -locking cancelled out the common fiber noise. Furthermore, we covered the fiber system with a box to avoid environmental fluctuations.

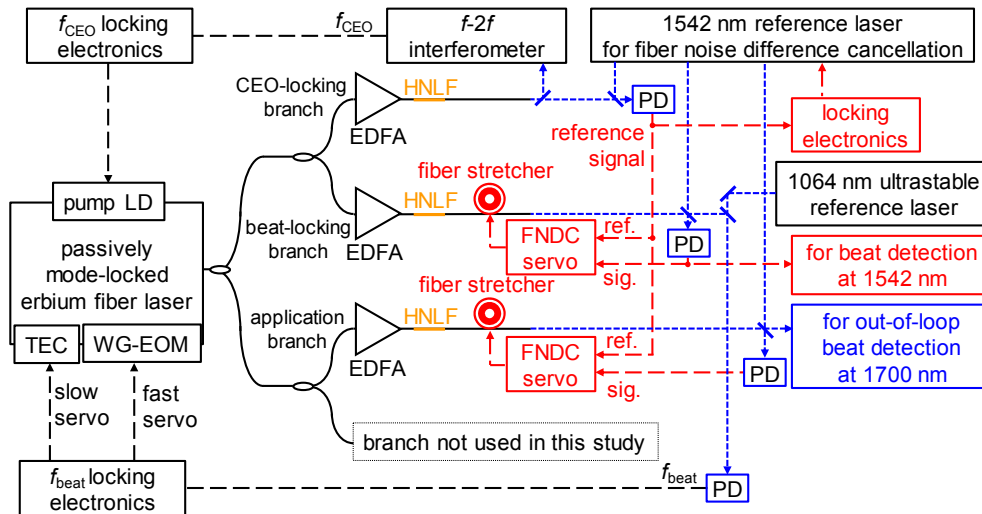


Fig. 1. Schematic diagram of experimental setup for reducing uncertainty in a multi-branch configuration. Black solid, and blue dotted lines denote optical fiber, and free-space parts, respectively. Black and red dashed lines are electronics parts for comb stabilization and fiber noise difference cancellation, respectively. TEC, temperature controller; WG-EOM, waveguide-type electro-optic phase modulator; LD, laser diode; EDFA, erbium-doped fiber amplifier; HNLf, highly nonlinear fiber; PD, photo detector; FNDC, (inter-branch) fiber noise difference cancellation.

2.2 Setup for comparison of two combs

To evaluate comb frequency stability, we prepared another quasi-identical multi-branch comb system. The relative frequency stability and the difference between the two combs were evaluated using the experimental setup shown in Fig. 2. The first and second comb setups described in subsection 2.1 are illustrated on the left (comb #1) and right (comb #2), respectively. The middle of Fig. 2 shows a setup for comparing the frequency stability of the two combs. We arranged the frequencies of the comb modes of the two combs as shown in Fig.

3. Suffixes “1” and “2” for f_{rep} , f_{CEO} , and f_{beat} describe the frequencies of comb #1 and #2, respectively. We controlled the oscillator cavity length of comb #2 using an intracavity delay line so that it had the same f_{rep} as comb #1. The two combs were referred to the common ultrastable reference laser at 1064 nm to stabilize f_{beat} and f_{rep} at 30 and 43.4 MHz, respectively. They had the same f_{beat} and f_{CEO} but in different signs at 30 MHz. ($f_{\text{beat}1} = -f_{\text{beat}2} = 30$ MHz, $f_{\text{CEO}1} = -f_{\text{CEO}2} = 30$ MHz, $f_{\text{rep}1} = f_{\text{rep}2} = 43.4$ MHz) The CW laser at 1542 nm was commonly used to detect reference and error signals in the CEO-locking branch and the other two branches for comb #2, respectively. The phase differences between the branches of comb #2 were cancelled out by the signals through the FNDC.

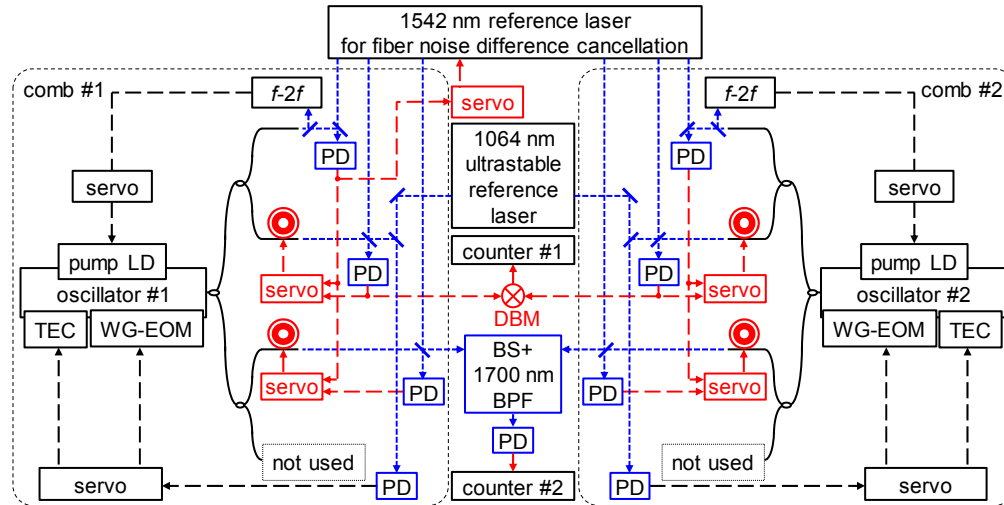


Fig. 2. Schematic diagram of experimental setup for optical comparison of two quasi-identical combs. BS, beam splitter; BPF, band pass filter; DBM, double balanced mixer.

The beat frequencies between the two combs were measured using beat-locking and application branch outputs. We counted the beat frequencies with π -type dead-time-free counters with a 1-s gate time. We used tracking oscillators for all the beat frequency measurements performed in this study. Here, we used two methods to obtain the beat signals. One was the direct beating between the two combs and the other was the signal mixing of the two beat signals between the FNDC reference laser and the each of the comb outputs. The signals obtained by the two methods have identical fluctuations. The former method requires the pulse overlapping of the two combs but the latter does not. In addition, it is difficult to overlap the pulses in two branches simultaneously, and we used the two different methods in the two respective branches. First, we measured the beat frequency of the beat-locking branch outputs from a mixer output at 60 MHz using counter #1. The 60 MHz signal was generated by mixing the beat notes between respective beat-locking branch outputs of the two combs and the FNDC reference laser at 1542 nm. Second, we measured the out-of-loop beat frequency at 1700 nm by measuring the direct beat at 60 MHz between the two comb outputs from the respective application branches. The two output beams were combined by a beam splitter and spectrally sliced by a band pass filter centered at 1700 nm with a 10 nm bandwidth. The two beams interfered at a PD and generated an out-of-loop beat signal at 60 MHz with a signal to noise ratio of ~ 45 dB at a 300 kHz resolution bandwidth. To overlap the pulses of the two combs in time for beat detection, we slightly changed the f_{rep} of comb #2 by tuning the reference frequency for the f_{CEO} or f_{beat} stabilization from 30 MHz and then returned it to 30 MHz when the out-of-loop beat signal appeared. The beat frequency between the application branch outputs was measured with counter #2.

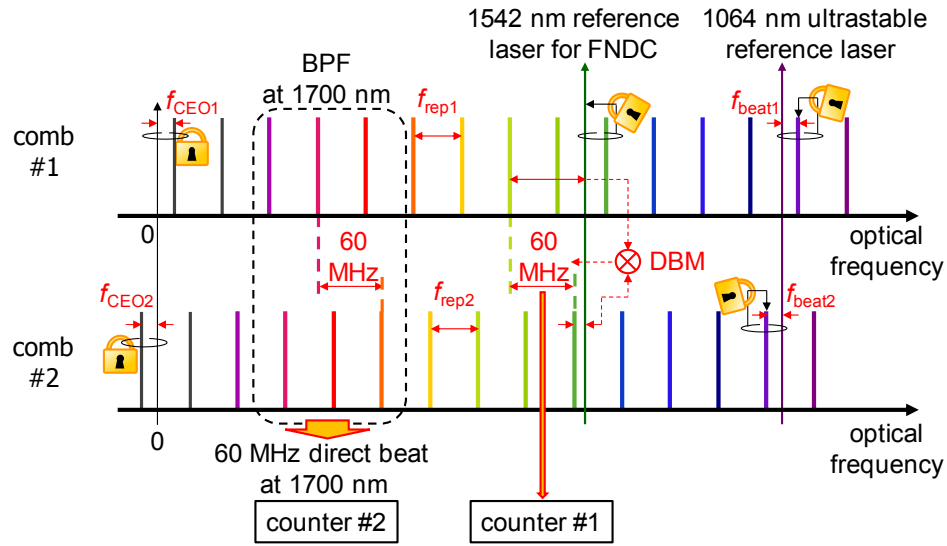


Fig. 3. Illustration of frequency arrangement of the two combs and the FNDC reference and ultrastable reference lasers.

3. Results and discussion

To confirm whether FNDC is effective in improving frequency stability, we turned it on during a measurement sequence. Figure 4 shows an example of such a measurement of the direct beat from two application branch outputs at 1700 nm. Once the FNDC had been turned on, the beat frequency distribution narrowed. This result indicates that FNDC improved the frequency stability of the comb from out-of-loop branches and the fiber noise differences between the branches constitute a major reason for the uncertainty degradation in the multi-branch fiber combs.

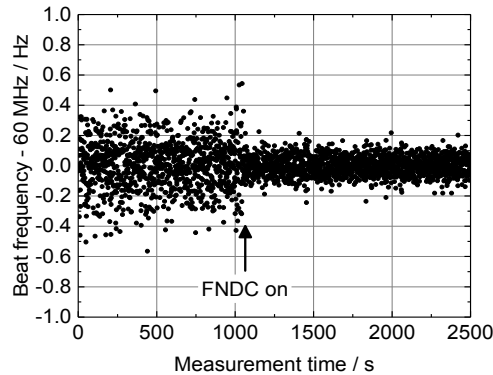


Fig. 4. Out-of-loop beat frequency between two combs from application branches with and without FNDC. The FNDC was turned on around a measurement time of 1000 sec. The averaging time was 1-s.

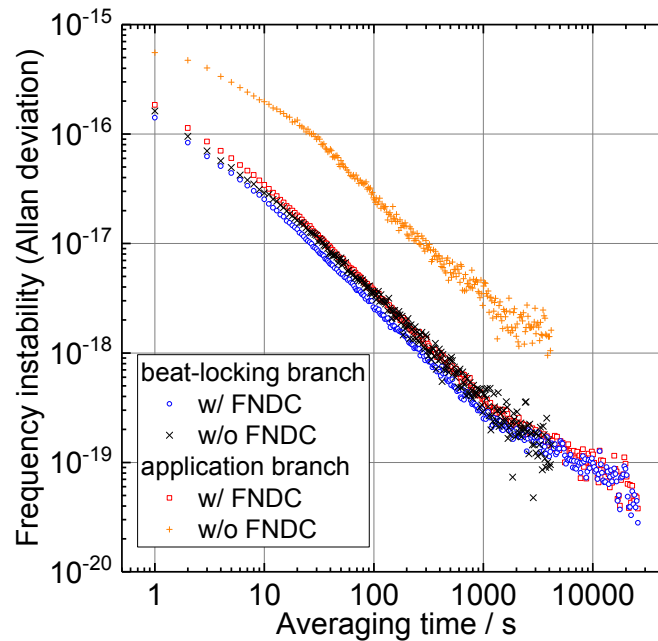


Fig. 5. Frequency instabilities given by Allan deviation with and without FNDC in beat-locking and application branches. All stability values are normalized to the carrier frequencies and are divided by the square root of 2 to obtain the stabilities for the individual combs.

The Allan deviation frequency instabilities were calculated from the recorded frequency sequences with and without the FNDC in beat-locking and application branches at 1542 and 1700 nm, respectively, as shown in Fig. 5. We calculated the Allan deviation values from frequency time series of 130000-s and 21000-s with and without FNDC, respectively. The best stability among the four results was obtained with FNDC measured from a beat-locking branch at 1542 nm. It was 1.4×10^{-16} at an averaging time of 1 s. The Allan deviation improved in almost inverse proportion to the averaging time and reached an order of 10^{-19} at an averaging time of ~ 10000 s. With the multi-branch configuration of the fiber-based combs, it is also important to evaluate the frequency instabilities of comb modes at a branch outside the comb stabilization loop. In the application branch, a frequency stability of 1.8×10^{-16} was obtained at an averaging time of 1 s and was improved to the order of 10^{-19} at an averaging time of ~ 10000 s, but this was slightly worse than the result measured in the beat-locking branch in all averaging times.

The frequency instabilities for a short averaging time measured in the beat-locking branch were better than those in the application branch both with and without FNDC. With the frequency instability measured using a beat-locking branch output, the fiber noise induced in the branch was partly cancelled out by the f_{beat} -locking servo even without the FNDC. On the other hand, the application branch output was completely out-of-loop for the comb stabilization, and the frequency stability measured by using this output corresponded to the evaluation of just the FNDC effect. Therefore, the FNDC greatly improved the frequency stability in the application branch. This branch is most important because it is freely available and can produce beat notes with a higher signal to noise ratio by spectral tailoring capability compared with the other branches or combs with single-branch/path configurations.

Here, we discuss what limits the frequency stability of multi-branch combs using the FNDC technique. In this study, we introduced fiber stretchers for the FNDC to the combs that we used in our previous report [12]. Figure 6 shows the power spectral density (PSD) of the phase noise from the out-of-loop direct beat at 1700 nm between the two combs without the FNDC. The

figure also shows Allan variance at a 1-s averaging time, which we calculated by applying a filter function to the phase noise PSD and integrated from 0.1 Hz toward higher frequency. To show the degree to which the spectral components of the phase noise influence the frequency instability, Fig. 6 shows the integrated Allan variance, not the Allan deviation. Although we did not measure the phase noise PSD with the FNDC, we can understand how the frequency stability was improved by the FNDC from Fig. 6 to some extent.

We consider the Allan deviation at a 1-s averaging time, which is commonly used to evaluate short-term frequency instabilities. First, in advance of the current study, we considered the Allan deviation at a 1-s averaging time as shown in our previous report [12]. We employed a tracking oscillator for the optical frequency measurement so that the phase noise in a frequency region higher than ~20 kHz had less effect on the measurement. The integrated Allan deviation at 20 kHz is 3×10^{-16} for the individual combs, which we calculated from the square root of an integrated Allan variance of 1.6×10^{-32} within the frequency region divided by the square root of 2 for the individual combs. The result agrees well with the result measured with the counter (3×10^{-16}) in the reference [12]. In addition, the frequency stability degraded when the phase-lock-loop (PLL) gain of the tracking oscillator increased. This gain increase expands the PLL bandwidth. These results indicate that the PLL bandwidth was around 20 kHz.

When the tracking oscillator was not used in the current and previous work, the Allan deviation was dominantly determined by the phase noise in a high frequency region of more than 30 kHz. Figure 6 shows that this phase noise component corresponds to an Allan deviation of about 1×10^{-15} . This result agrees with our measurement result obtained with a frequency counter (not shown). The phase noise in the high frequency range originated from noise floor in the beat spectrum where the shot noise was dominant. The noise component can be suppressed by narrowing the measurement bandwidth, for example, by introducing a tracking oscillator. This can improve the Allan deviation at a 1-s averaging time.

Second, we consider the Allan deviation at a 1-s averaging time in this study. With a short averaging time, the relative instability is slightly worse than our previous result. In the previous report, we covered all the branches with a box to suppress environmental noise. In the current study, we introduced PZT-based fiber stretchers in the two branches for the FNDC, and these fibers could not be fully covered because of the cabling. We assume that the increase in the fiber lengths of the branches and the branch enclosure problem degraded the frequency stability mainly due to the environmental noise. This noise may increase the phase noise in a low frequency region of less than 1 kHz. This additional phase noise might increase the Allan deviation to about 6×10^{-16} in the application branch without the FNDC (Fig. 5).

The FNDC can suppress the phase noise in the low frequency region. This was realized by employing fiber length control with the drum-type PZT, and its control bandwidth was about 100 Hz. The FNDC suppressed the phase noise at a low frequency of less than 100 Hz, resulting in $\sim 2 \times 10^{-16}$ frequency instability in the application branch with the FNDC (Fig. 5). The integrated Allan deviation from 100 Hz to 20 kHz in Fig. 6 was 1.2×10^{-16} for a single comb, which we calculated from an integrated Allan variance of 3×10^{-32} within the frequency region. This discrepancy may be due to the phase noise increase originating from the fiber length increase of the branches and the branch enclosure problem.

In this study, some part of the optical path length fluctuation was not cancelled; the f_{CEO} beat detection path and part of the beat detection path for the FNDC. Although the independent optical path fluctuations remained in each branch, these parts are all composed in free space; we assume that they do not induce frequency instability of 10^{-16} order with a 1-s averaging time. It is difficult to distinguish the frequency instabilities induced by these optical path lengths fluctuations and remaining inter-branch fiber noise. A further investigation is required to understand the origin of the frequency instability limitation.

Summarizing the above discussion, the phase noises in frequency regions lower and higher than 100 Hz and 20 kHz were suppressed by the FNDC and the tracking oscillator, respectively. Thus, the remaining phase noise between 100 Hz and 20 kHz was dominant as regards the

frequency instability. The Allan deviation at 1-s averaging with the FNDC and the tracking oscillator was mainly limited by the power supply noise in the 50 Hz to 1 kHz range. The Allan deviation induced by the noise is estimated to be 8×10^{-17} . This noise was assumed to limit the short-term frequency stability and can be reduced by some technical improvement.

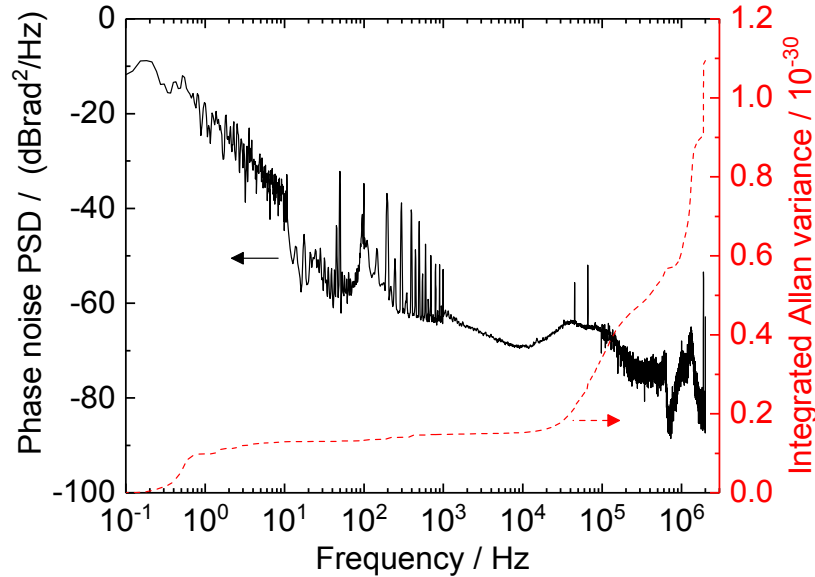


Fig. 6. Power spectral density of phase noise of the out-of-loop beat at 1700 nm (black solid line) [12] and integrated Allan variance (red dashed line) without FNDC. The Allan variance is calculated from the phase noise PSD. The Allan variance is normalized by a wavelength of 1700 nm.

Another concern is the frequency instability with the long averaging time. When we consider that the FNDC had sufficiently suppressed the low frequency phase noise, the limitation will be optical path length fluctuation of the uncanceled free-space parts caused by the environmental change. Williams et al. reported that the frequency instability induced by a covered free space optical path tens of cm in length was $2 \times 10^{-17} / \tau^{1/2}$ in modified Allan deviation where τ is the averaging time [33]. Although we have discussed the frequency instability in terms of Allan deviation, we can simply compare the values in a long averaging time region where white noise has less influence on frequency stability. According to the report, the frequency instability induced by the free space path is around 6×10^{-19} at a 1000-s averaging time. In our experiment, the free-space optical path lengths which were outside the FNDC servo loop were similar to the lengths in the report. Consequently, the Allan deviation was scaled down slower by $\tau^{-1/2}$ at longer than 1000 sec in Fig. 5.

We measured the long-term frequency instability more than 30 times but did not obtain a value of less than 10^{-20} . To reach a frequency uncertainty of 10^{-21} order, further improvement is required, such as shortening the out-of-loop path length and reducing the power supply noise.

To reduce the effect of white phase noise, we analyzed the frequency instabilities by calculating the modified Allan deviation as shown in Fig. 7. In this case, the averaging time and the total measurement time were 100 s and 510000 s, respectively. The frequency instabilities reached a sub- 10^{-20} order (9.7×10^{-21}) at an averaging time of 126500 s. In addition, we calculated the moving average series of the frequency time series from the 600000-s measurement. The averaging time was from 100 to 300000 s. We calculated the weighted mean and the weighted standard deviation from the moving average series to be 1.9×10^{-20} and 2.3×10^{-20} , respectively. By dividing the weighted standard deviation by the square root of 2, we obtained a weighted standard deviation for a single comb of 1.6×10^{-20} . These results

suggested that we achieved a practical multi-branch fiber-comb with a relative uncertainty at the 10^{-20} level.

The FNDC can sufficiently suppress the inter-branch fiber noise difference and can reduce the frequency stability by one order from shorter and longer averaging times. However, some optical path fluctuation remains uncanceled and, therefore, phase noise also remains to some extent in the low frequency region. A drawback of the FNDC technique is that the whole system is more complicated because of the additional beat detection and servo control setups.

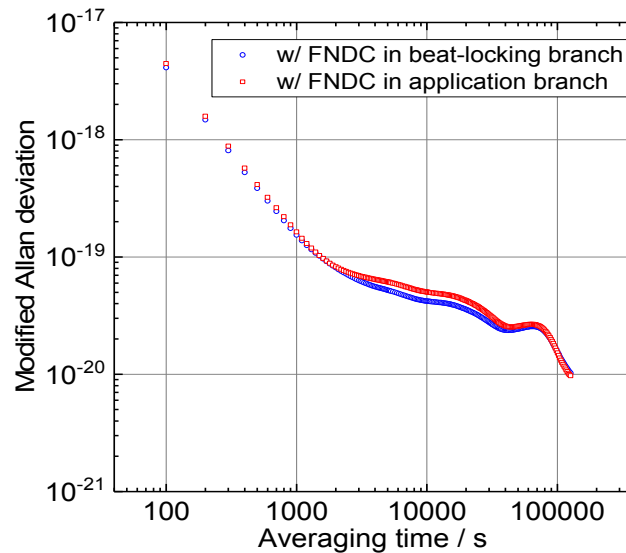


Fig. 7. Frequency instabilities given by the modified Allan deviation with FNDC in beat-locking and application branches. All stability values are normalized to the carrier frequencies and are divided by the square root of 2 to obtain stabilities for the individual combs.

4. Conclusion

We introduced FNDC in the respective branches of a multi-branch fiber comb to achieve high frequency stability. Two quasi-identical combs were constructed for a frequency stability comparison in the optical domain. We evaluated the frequency stabilities of the combs from two branches with FNDC at different wavelengths. The FNDC enhanced the stability of the comb and enabled us to achieve 1.4×10^{-16} and 1.8×10^{-16} at a 1-s averaging time in beat-locking and application branches, respectively. Finally, the longer averaging time improved the frequency stabilities in both branches. A frequency stability slightly better than 10^{-20} was achieved in modified Allan deviation. Moreover, we obtained a weighted standard deviation for a single comb of 1.6×10^{-20} . The fiber combs offer robustness, reliability, and long-term stability thanks to their all fiber-based design. The multi-branch configuration enabled us to control spectral profiles thus enhancing the high signal to noise ratio for adaptation to applications of each branch output. This study described a technique for overcoming a drawback of the multi-branch configuration, namely the frequency stability degradation caused by phase noise differences among branches.

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